

Lunar Surface Analysis via XRF

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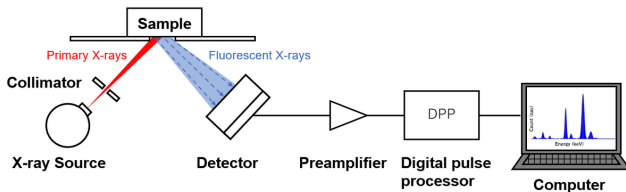
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- **Goal:** Identification and quantification of chemical elements on the lunar surface.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of multiel-spectra for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, different for each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.



The analysis focuses on determining the chemical elements within the XRF spectra through three interconnected computational tasks:

- 1 Classification
- 2 Regression
- 3 Denoising

To train models, we developed a Generator that simulates physical variations in XRF spectra.

- **Composition:** Samples contain **0 to 5** elements randomly selected from **41** possible candidates.
- **Stochasticity:** To ensure model robustness, each spectrum is generated by randomly varying operating conditions and noise levels simulating real-world measurement variability.

The classification task focus on identifying the elemental composition by analyzing spectral signatures.

Analytical stages

- **Multi-label classification:** across a pool of 41 potential elements.
- **Multi-class classification:** across 5 classes (1 to 5 elements) defined by the number of elements present in the sample.

Preprocessing Strategies and Architectures

Preprocessing Scaling

Tested to handle high dynamic range in spectral counts:

- **Standard:** 0 mean and 1 variance.
- **MinMax:** Rescaling to $[0, 1]$ range.
- **Log-MinMax:** Logarithmic transformation to compress peaks.

Model Architectures

Machine Learning:

- Random Forest, AdaBoost, XGBoost

Deep Learning:

- MLP, CNN, Transformer

Results: Example Multi-label Classification

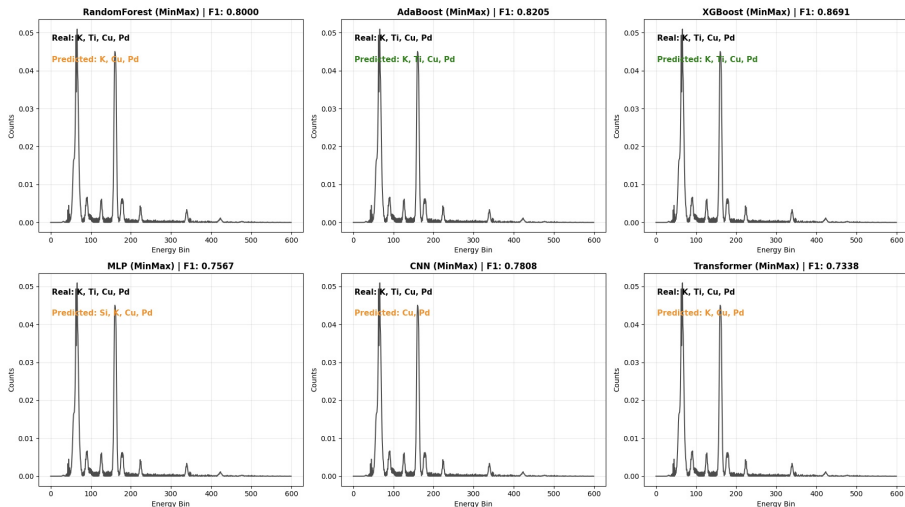


Figure: Predicted vs. real elemental composition for a sample

Results Multi-label Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
Standard	XGBoost	0.8691	0.9593	0.8237
MinMax	XGBoost	0.8691	0.9589	0.8243
LogMinMax	XGBoost	0.8691	0.9589	0.8243
MinMax	AdaBoost	0.8205	0.9151	0.7821
LogMinMax	AdaBoost	0.8203	0.9088	0.7830
Standard	AdaBoost	0.8142	0.9049	0.7770
LogMinMax	RandomForest	0.8004	0.9618	0.7314
MinMax	RandomForest	0.8000	0.9600	0.7305
Standard	RandomForest	0.7995	0.9848	0.7292
LogMinMax	CNN	0.7979	0.9719	0.7267
Standard	CNN	0.7910	0.9597	0.7203
MinMax	CNN	0.7808	0.9567	0.7062
MinMax	MLP	0.7567	0.9419	0.6802
LogMinMax	MLP	0.7542	0.9494	0.6751
MinMax	Transformer	0.7338	0.9441	0.6540
Standard	MLP	0.7320	0.9351	0.6415
Standard	Transformer	0.7173	0.8914	0.6331
LogMinMax	Transformer	0.7141	0.9299	0.6295

Figure: Multi-label classification results ranked by F1-Score.

Results Multi-class Classification

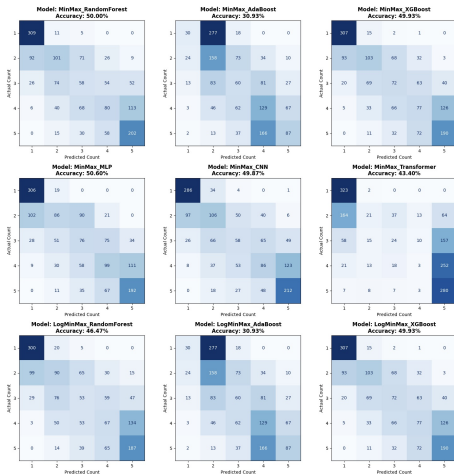


Figure: Confusion matrices for multi-class classification.

Results: Example Multi-class Classification

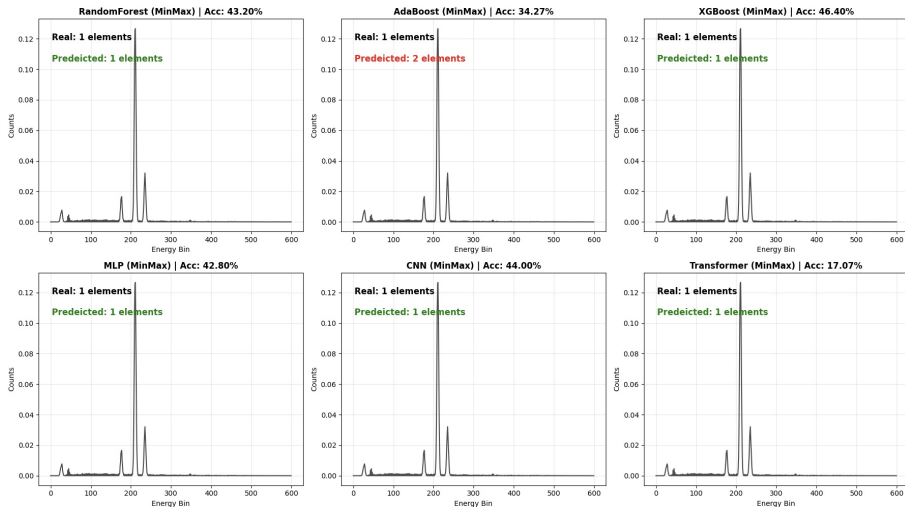


Figure: Predicted vs. real number of elements.

Results Multi-class Classification

Preprocessing	Architecture	F1-Score	Precision	Recall
LogMinMax	XGBoost	0.4473	0.4411	0.4653
MinMax	XGBoost	0.4467	0.4402	0.4640
Standard	RandomForest	0.4288	0.4313	0.4440
Standard	XGBoost	0.4240	0.4203	0.4413
MinMax	CNN	0.4207	0.4330	0.4400
LogMinMax	CNN	0.4185	0.4230	0.4307
MinMax	RandomForest	0.4137	0.4148	0.4320
MinMax	MLP	0.4023	0.4007	0.4280
LogMinMax	RandomForest	0.3820	0.3780	0.4027
LogMinMax	MLP	0.3686	0.3786	0.4133
Standard	CNN	0.3650	0.3579	0.3853
Standard	AdaBoost	0.3327	0.3489	0.3427
MinMax	AdaBoost	0.3327	0.3489	0.3427
LogMinMax	AdaBoost	0.3327	0.3489	0.3427
Standard	MLP	0.2388	0.2694	0.3013
Standard	Transformer	0.0802	0.0490	0.2213
LogMinMax	Transformer	0.0776	0.0472	0.2173
MinMax	Transformer	0.0498	0.0291	0.1707

Figure: Multi-class classification results ranked by F1-Score.